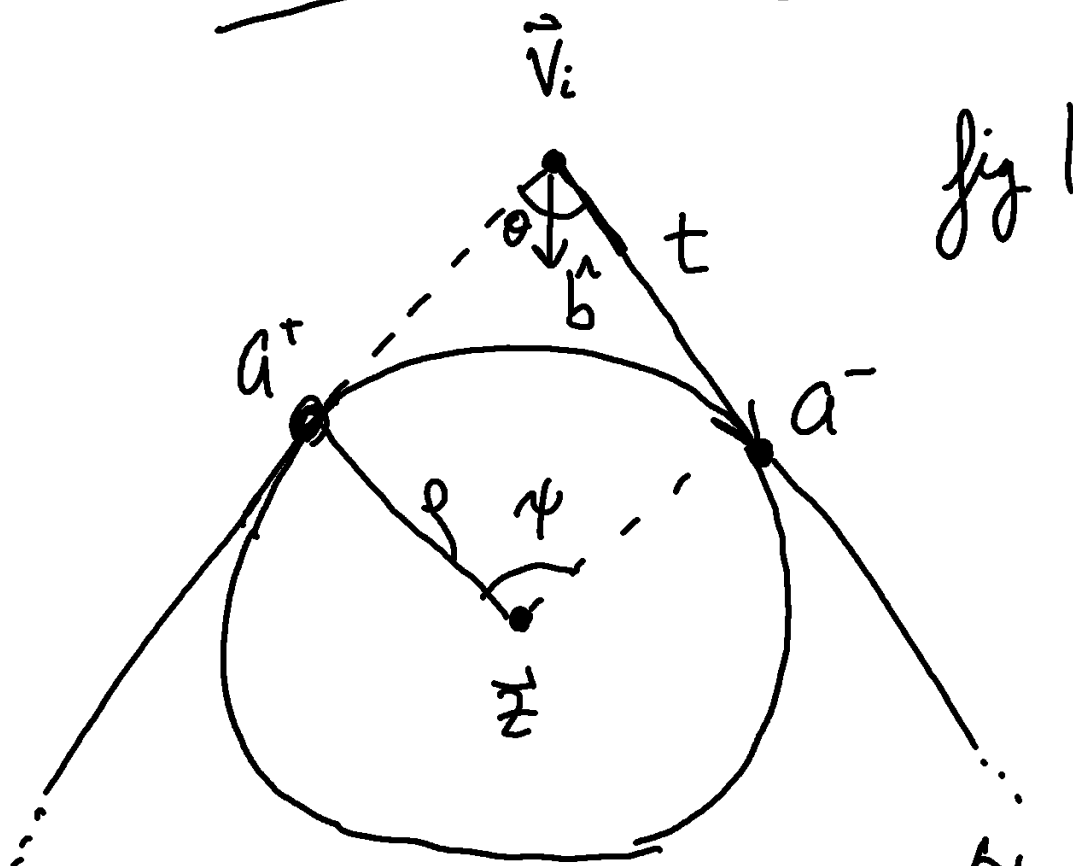


1. Rounded Definitions



$$\vec{a}_i^- = \vec{V}_i - t_i \hat{e}_{z'(i)}$$

$$\vec{a}_i^+ = \vec{V}_i + t_i \hat{e}_i \quad 1.1$$

$$\theta_i = \arccos(-\hat{e}_{z'(i)} \cdot \hat{e}_i) \quad 1.4$$

$$\hat{b}_i = \frac{\hat{e}_i - \hat{e}_{z'(i)}}{|\hat{e}_i - \hat{e}_{z'(i)}|} \quad 1.5$$

$$\tan \theta/2 = \frac{p}{t}$$

$$t_k = \frac{p}{\tan \theta_k/2} \quad 1.2$$

$$\sin \theta/2 = \frac{p}{|\vec{V}_i - \vec{z}|}$$

so

$$\vec{z}_i = \vec{V}_i + \left(\frac{p}{\sin \theta_i} \right) \hat{b}_i \quad 1.3$$

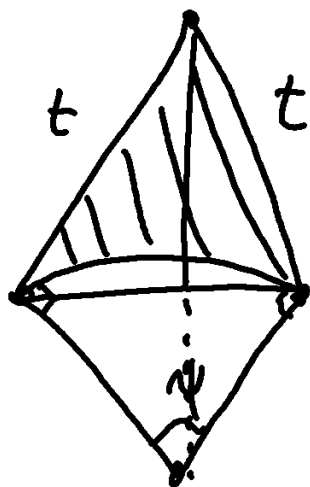
$$\psi_i = \pi - \theta_i \quad 1.6$$

Let's define (I'll surely run out of letters)

$$C_i \equiv \frac{\rho}{\sin(\theta_i/2)} \quad 1.7$$

so

$$\vec{z}_i = \vec{v}_i + C_i \hat{b}_i \quad 1.8$$



$$\Delta p = -2t + \rho\psi \quad 1.9$$

Find the area of the backlobe now with a line from a to a' and add only the circular segment.

$$\Delta A = \frac{1}{2} \psi \rho^2 - \frac{1}{2} \rho^2 \sin \psi$$

$$\Delta A = \frac{1}{2} \rho^2 (\psi - \sin \psi) \quad 1.10$$

Now the gradient

$$\begin{aligned} \frac{2\theta_k}{2V_j^\alpha} &= \frac{2}{2V_j^\alpha} \arccos \left[-\hat{e}_{z'(k)} \cdot \hat{e}_k \right] \\ &= \frac{(\delta_{kj} - \delta_{z'(k)j}) \left(\frac{\hat{e}_k^\alpha - (\hat{e}_k \cdot \hat{e}_{z'(k)}) \hat{e}_{z'(k)}^\alpha}{|\hat{e}_{z'(k)}|} \right)}{\sin \theta_k} \\ &\quad + \frac{(\delta_{z'(k)j} - \delta_{kj}) \left(\frac{\hat{e}_{z'(k)}^\alpha - (\hat{e}_k \cdot \hat{e}_{z'(k)}) \hat{e}_k^\alpha}{|\hat{e}_k|} \right)}{\sin \theta_k} \end{aligned}$$

$$\begin{aligned} \frac{2\theta_k}{2V_j^\alpha} &= \frac{(\delta_{kj} - \delta_{z'(k)j})}{\sin \theta_k} \left(\frac{\hat{e}_k^\alpha - \cos \theta_k \hat{e}_{z'(k)}^\alpha}{|\hat{e}_{z'(k)}|} \right) \\ &\quad + \frac{(\delta_{z'(k)j} - \delta_{kj})}{\sin \theta_k} \left(\frac{\hat{e}_{z'(k)}^\alpha - \cos \theta_k \hat{e}_k^\alpha}{|\hat{e}_k|} \right) \end{aligned}$$

1.11.

$$P = P_{\text{backbone}} - \sum_{j=1}^n (2t_j - p \psi_j)$$

$$\begin{aligned} \frac{2P}{2V_k^\alpha} &= \sum_{j=1}^n \frac{2}{2V_k^\alpha} \left[|\vec{V}_{z(j)} - \vec{V}_j| - 2t_j + p \psi_j \right] \\ &= \sum_{j=1}^n \left[\hat{e}_j^\alpha (\delta_{z(j)k} - \delta_{jk}) - 2 \frac{2}{2V_k^\alpha} \frac{p}{\tan \theta_{j/2}} \right. \\ &\quad \left. - p \frac{2\theta_j}{2V_k^\alpha} \right] \end{aligned}$$

$$\frac{2P}{2V_k^\alpha} = \hat{e}_{z'(k)}^\alpha - \hat{e}_k^\alpha + p \sum_{j=1}^n \frac{2\theta_j}{2V_k^\alpha} (-1 + \csc^2 \theta_{j/2})$$

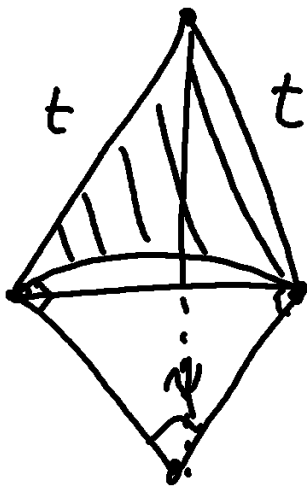
$$\boxed{\frac{2P}{2V_k^\alpha} = \hat{e}_{z'(k)}^\alpha - \hat{e}_k^\alpha + p \sum_{i=1}^n \frac{2\theta_i}{2V_k^\alpha} \cot^2 \theta_{i/2}} \quad 1.12 \quad \bullet$$

Next we separate this into a kite, wedge, and polygon

$$A = A_{\text{backbone}} - \sum_{j=1}^n (A_{\text{kite},j} - A_{\text{wedge}}) S_j$$

Convexity $\nearrow$

$$= \sum_{j=1}^n \left[\frac{1}{2} \epsilon^\alpha_\beta e^\beta_j e^\alpha_{z(j)} - A_{\text{kite},j} + \frac{\psi_j}{2} \rho^2 \right]$$



$$A_{\text{kite}} = \frac{1}{2} (2t \sin \theta/2) (p \cos \psi/2 + t \cos \theta/2)$$

$$= t \sin \theta/2 \left(p \frac{\sin^2 \theta/2 + \cos^2 \theta/2}{\sin \theta/2} \right)$$

$$A_{\text{kite}} = tp$$

$$A_{\text{kite}} = \frac{p^2}{\tan \theta/2} \quad 1.13$$

$$A_{\text{kite}} - A_{\text{wedge}} = p^2 (\cot \theta/2 - \psi/2) \quad 1.14$$

$$\frac{2A}{2V_{1c}^\alpha} = \frac{1}{2} \epsilon^\alpha_\beta (\vec{V}_{z(1c)} - \vec{V}_{z'(1c)})^\beta + \frac{1}{2} p^2 \sum_{i=1}^n S_i \cot^2 \left(\frac{\theta_i}{2} \right) \frac{2\theta_i}{2V_{1c}^\alpha}$$

1.15

$\uparrow$ from 1-csc²

Intersections

$$\text{Let } \boxed{\vec{P}_k(s) = \vec{a}_k^+ + s(\vec{a}_{z(k)}^- - \vec{a}_k^+)} \quad 1.16$$

be a straight segment where $s \in [0, 1]$.

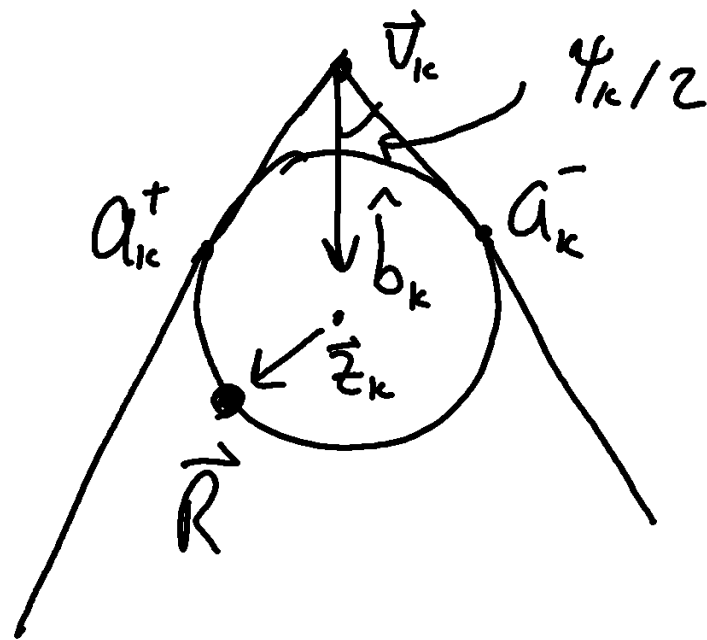
Points on the arc have the form

$$\boxed{\vec{Q}_k(\varphi) = \vec{z}_k + \rho_k \begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix}} \quad 1.17$$

and the arc is defined for

$$\boxed{\alpha_k^\pm = \arctan(\vec{a}_k^\pm - \vec{z}_k)} \quad 1.18$$

and spans $\psi_k = \alpha_k^+ - \alpha_k^-$.



For a point $\vec{R}$ on the circle,

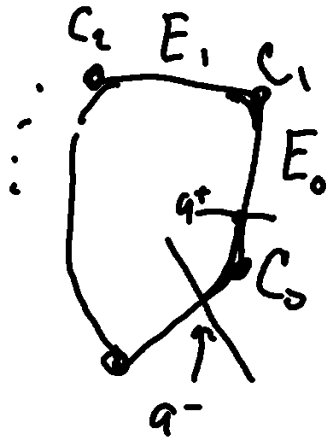
$\vec{R}$ lies on the arc iff

$$\frac{\vec{R} - \vec{z}_k}{p_k} \cdot \hat{b}_k \leq -\cos \psi_k/2 \quad (AVT) \quad 1.19$$

Coin this AVT for arc-validity test.

2. Ordering features

We order the features of a polygon like



so there are $2n$ features per polygon

The crossing value is (where A and B are the intersecting pair)

$$\sigma_A = f + \begin{cases} \lambda & \text{if } f \text{ even} \\ s & \text{if } f \text{ odd} \end{cases} \quad 2.1$$

Where λ is the fractional part $\lambda \in [0, 1)$ of the arc. For an arc $k = f/2$, we have

$$2.2 \quad \lambda \equiv \frac{\alpha - \alpha_k^-}{\psi_k} \quad \text{since } \alpha_k^+ - \alpha_k^- = \psi_k$$

We need to find α (where the intersection occurs) and s .

finding S

S depends on the sort of intersection you have.

If you have two edges, then

$$\begin{aligned}\vec{P}_k(s_1) &= \vec{a}_k^+ + s_1 \vec{u}_k, & \vec{u}_k &= \vec{a}_{\tilde{z}(k)} - \vec{a}_k^+ \\ \vec{P}_k(s_2) &= \vec{a}_\ell^+ + s_2 \vec{u}_\ell\end{aligned}\quad 2.3$$

if we let $\vec{w}_{k\ell} = \vec{a}_\ell^+ - \vec{a}_k^+$ then

$$\vec{P}_\ell(t) - \vec{P}_k(s) = \vec{w}_{k\ell} - s_1 \vec{u}_k + s_2 \vec{u}_\ell = \vec{0} \text{ for}$$

an intersection. Therefore

$$-s_1 u_k^x + s_2 u_\ell^x + w_{k\ell}^x = 0$$

$$s_2 = \frac{s_1 u_k^x - w_{k\ell}^x}{u_\ell^x}$$

$$-S_1 U_{1k}^y + \left(\frac{S_1 U_{1k}^x - W_{k\ell}^x}{U_{\ell}^x} \right) U_{\ell}^y + W_{k\ell}^y = 0$$

$$S_1 \left(-U_{1k}^y + \frac{U_{1k}^x}{U_{\ell}^x} U_{\ell}^y \right) = \frac{W_{k\ell}^x}{U_{\ell}^x} U_{\ell}^y - W_{k\ell}^y$$

$$S_1 = \frac{W_{k\ell}^x U_{\ell}^y - W_{k\ell}^y U_{\ell}^x}{-U_{1k}^y U_{\ell}^x + U_{1k}^x U_{\ell}^y}$$

$$S_1 = \frac{\epsilon_{\alpha\beta} W_{k\ell}^{\alpha} U_{\ell}^{\beta}}{\epsilon_{\alpha\beta} U_{1k}^{\alpha} U_{\ell}^{\beta}} \quad 2.4$$

If the edge intersects an arc then

$\vec{P}_{1k}(s)$ intersects a point $\vec{R}_{\ell}$ on the full circle associated with arc ℓ . Then $|\vec{R}_{\ell} - \vec{z}_{\ell}| = r$

If we then let $\vec{W}_{k\ell} = \vec{a}_k^+ - \vec{z}_{\ell}$ where the segment starts at point $\vec{a}_k^+$, then we can proceed.

this gives $\vec{R}_\ell = \vec{P}_k(s)$ so that

$$|\vec{R}_\ell - \vec{z}_\ell|^2 = |\vec{P}_k(s) - \vec{z}_\ell|^2$$

$$\begin{aligned} \rho^2 &= |\vec{w}_{k\ell} + S_1 \vec{u}_k|^2 \\ &= |\vec{w}_{k\ell}|^2 + 2S_1(\vec{w}_{k\ell} \cdot \vec{u}_k) + S_1^2 |\vec{u}_k|^2 \end{aligned}$$

Then

$$0 = \left(\frac{|\vec{w}_{k\ell}|}{|\vec{u}_k|} \right)^2 + 2S_1 \frac{\vec{w}_{k\ell} \cdot \vec{u}_k}{|\vec{u}_k|^2} + S_1^2 - \rho^2$$

$$S_1^\pm = - \frac{\vec{w}_{k\ell} \cdot \vec{u}_k}{|\vec{u}_k|^2} \pm \frac{1}{|\vec{u}_k|^2} \sqrt{(\vec{w}_{k\ell} \cdot \vec{u}_k)^2 + |\vec{u}_k|^2 (\rho^2 - |\vec{w}_{k\ell}|^2)}$$

$$\text{Let } D_{k\ell} = (\vec{w}_{k\ell} \cdot \vec{u}_k)^2 + |\vec{u}_k|^2 (\rho^2 - |\vec{w}_{k\ell}|^2) \quad 2.5$$

$$S_1^\pm = \frac{-(\vec{w}_{k\ell} \cdot \vec{u}_k) \pm \sqrt{D_{k\ell}}}{|\vec{u}_k|^2} \quad 2.6$$

Keep each root if $S_i^\pm$ is valid
(between 0 and 1 and passes AVT).

It could be that an edge intersects an arc twice.

To get the corresponding angle α , we simply find $\vec{P}_k(s)$ and then have

$$\alpha_k = \arccos \left(- \frac{(\vec{P}_k(s_i) - \vec{z}_{ic}) \cdot \vec{b}_k}{\rho_k} \right) \quad 2.7$$

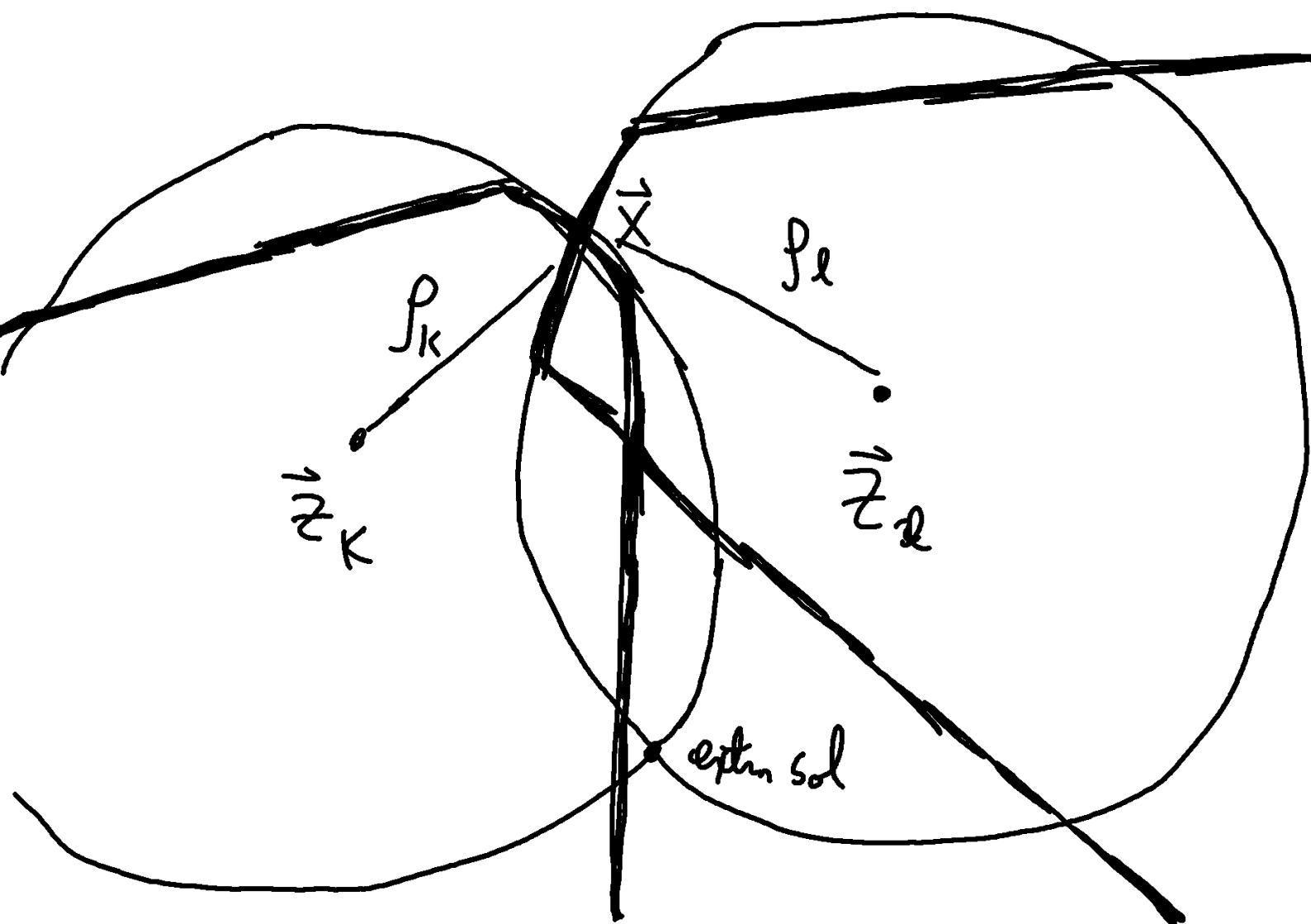
3. Arc-Arc Intersection

We consider two arcs colliding at point

$\vec{X}$ so that $|\vec{X} - \vec{z}_k|^2 = \rho_k^2$ and

$$|\vec{X} - \vec{z}_l|^2 = \rho_l^2$$

These solutions should lie on a line



Solving,

$$|\vec{x}|^2 - 2\vec{x} \cdot \vec{z}_k + |\vec{z}_k|^2 = \rho_k^2$$

$$|\vec{x}|^2 - 2\vec{x} \cdot \vec{z}_l + |\vec{z}_l|^2 = \rho_l^2$$

$$2\vec{x} \cdot (\vec{z}_l - \vec{z}_k) + |\vec{z}_k|^2 - |\vec{z}_l|^2 = \rho_k^2 - \rho_l^2$$

if we let $\vec{n}_{kl} = \vec{z}_l - \vec{z}_k$ and

$$\hat{n}_{kl} = \frac{\vec{n}_{kl}}{|\vec{n}_{kl}|} \quad \text{then}$$

$$\vec{x} \cdot \hat{n}_{kl} = \frac{\rho_k^2 - \rho_l^2 - |\vec{z}_k|^2 + |\vec{z}_l|^2}{2|\vec{z}_l - \vec{z}_k|}$$

Next we write

$$3.1 \quad \boxed{\vec{x}_{\pm} = \vec{z}_k + a_1 \hat{n}_{kl} \pm a_2 \hat{n}_{kl}^{\perp}}$$

so that

$$\begin{aligned} \vec{x}_{\pm} \cdot \hat{n}_{kl} &= \vec{z}_k \cdot \hat{n}_{kl} + a_1 \\ &= \frac{2(\vec{z}_k \cdot \vec{z}_l - |\vec{z}_k|^2)}{2|\vec{z}_l - \vec{z}_k|} + a_1 \end{aligned}$$

on either side of the middle

$$a_1 = \frac{p_k^2 - p_e^2 + |\vec{z}_k|^2 + |\vec{z}_e|^2 - 2\vec{z}_k \cdot \vec{z}_e}{2|\vec{z}_e - \vec{z}_k|}$$

$$a_1 = \frac{p_k^2 - p_e^2}{2|\vec{z}_e - \vec{z}_k|} + \frac{1}{2}|\vec{z}_e - \vec{z}_k| \quad 3.2$$

Since $|\vec{x}_\pm - \vec{z}_k|^2 = p_k^2$, we have

$$a_2 = \sqrt{p_k^2 - a_1^2}, \quad 3.3$$

4 Area overlap

Let polygons A and B intersect and generate a new polygon $A \cap B$ which has boundary $2(A \cap B)$.

The area of $A \cap B$ is

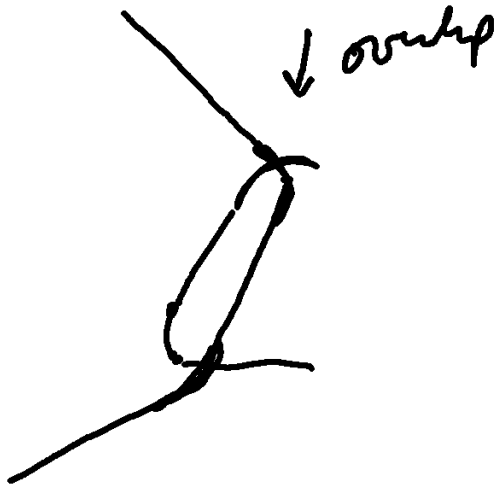
4.1

$$|A \cap B| \equiv \frac{1}{2} \epsilon_{\alpha\beta} \oint_{\partial(A \cap B)} x^\alpha dx^\beta.$$

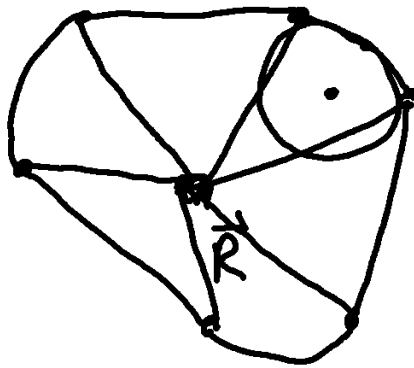
The area over an edge feature from point $\vec{P}$ to $\vec{Q}$ is

$$h_e(\vec{P}, \vec{Q}) \equiv \frac{1}{2} \epsilon_{\alpha\beta} P^\alpha Q^\beta$$

For the arc parts, the story is different



If I think about it, it shouldn't matter what type of crossing it is, only what the boundary feature is



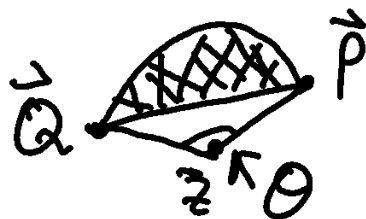
Where the reference point is $\vec{R}$.

For an overlap, we get the reference point as the first backbone vertex of polygon A given $A < B$.

Our edge-feature area component is

$$h_e(\vec{P}, \vec{Q}) \equiv \frac{1}{2} \epsilon_{\alpha\beta} (P^\alpha - R^\alpha)(Q^\beta - R^\beta) \quad 4.2$$

For the arc, you can treat $\vec{P}$ and $\vec{Q}$ as the points and separate the feature into a triangle and circular segment



We can find the sector angle θ
 And the area will be the sector minus
 the triangle formed by $\vec{z}$, $\vec{p}$, and $\vec{q}$.

$$\tan \theta = \frac{\epsilon_{\alpha\beta} (p^\alpha - z^\alpha)(q^\beta - z^\beta)}{(\vec{p} - \vec{z}) \cdot (\vec{q} - \vec{z})} \quad 4.3$$

Then

$$h_a(\vec{p}, \vec{q}) = \frac{1}{2} p^2 (\theta - \sin \theta) \quad 4.4$$

We can let

$$h(\vec{p}, \vec{q}) \equiv \begin{cases} h_a(\vec{p}, \vec{q}) & \text{if odd} \\ h_e(\vec{p}, \vec{q}) + h_a(\vec{p}, \vec{q}) & \text{if even.} \end{cases} \quad 4.5$$

Then

$$h(\vec{p}, \vec{q}) = \frac{1}{2} p^2 (\theta - \sin \theta)$$

Now we traverse the features.

We let the intersection pairs be P . There are the same number of P as intersections.

Let $P = \{i, j, k, l\}$ where feature i intersects feature j and following i CCW eventually leads to feature l which intersects feature k . To reiterate, the shapes are $\{A, B, B, A\}$.

$$U = \sum_{P=\{i,j,k,l\}} \left\{ \delta_{[i]=l} h(\vec{q}_i, \vec{q}_l) + (1 - \delta_{[i]=l}) \left[h(\vec{q}_i, \vec{q}_{z(l,i)}) + h(\vec{q}_{z(l,i)}, \vec{q}_f) + \sum_{m=z(l,i)}^{z'(l,i)} h(\vec{q}_m, \vec{q}_{z(m)}) \right] \right\}.$$

4.6

This sum is problematic for parallelization. We can separate this double sum piece out:

$$U_{out} = \sum_{p=\{i,j,k,l\}} \left[\delta_{i-l} h(\vec{q}_i, \vec{q}_l) + (1 - \delta_{i-l}) (h(\vec{q}_i, \vec{q}_{z(i)}) + h(\vec{q}_l, \vec{q}_f)) \right] \quad 4.7$$

$$U_{in} = \sum_{m=0}^{2N} \sum_{p=\{i,j,k,l\}} \delta_{m \in \{i, z(l)\}} h(\vec{q}_m, \vec{q}_{z(m)}) \quad 4.8$$

We can distribute the sum over threads and use a sorted set of features to efficiently search over pairs of intersections.

50 Overlap Forces

There are a lot of pieces $h(\vec{P}, \vec{Q})$.

To differentiate,

$$5.1 \quad \frac{\partial h}{\partial V_k^\alpha} = \left(\frac{\partial h_e}{\partial P^\beta} \right) \left(\frac{\partial P^\beta}{\partial V_k^\alpha} \right) + \left(\frac{\partial h_e}{\partial Q^\beta} \right) \left(\frac{\partial Q^\beta}{\partial V_k^\alpha} \right) + \delta_{\text{arc}} \left(\frac{\partial h_a}{\partial Z_k^\beta} \right) \left(\frac{\partial Z_k^\beta}{\partial V_k^\alpha} \right).$$

$$5.2 \quad \begin{aligned} \frac{\partial h_e}{\partial P^\gamma} &= \frac{1}{2} \epsilon_{\gamma\beta} (Q^\beta - R^\beta) \\ \frac{\partial h_e}{\partial Q^\gamma} &= -\frac{1}{2} \epsilon_{\gamma\beta} (P^\beta - R^\beta) \end{aligned}$$

Now we consider

$$\frac{\partial h_a}{\partial p^x} = \frac{1}{2} p^2 (1 - \cos \theta) \frac{\partial \theta}{\partial p^x}.$$

We can implicitly differentiate

$$\sec^2 \theta \frac{\partial \theta}{\partial p^x} = \frac{\partial}{\partial p^x} \left[\frac{\epsilon_{\alpha\beta} (p^\alpha - z^\alpha) (q^\beta - z^\beta)}{(\vec{p} - \vec{z}) \cdot (\vec{q} - \vec{z})} \right]$$

$$\sec^2 \theta \frac{\partial \theta}{\partial p^x} = \frac{(\vec{p} - \vec{z}) \cdot (\vec{q} - \vec{z}) \epsilon_{x\beta} (q^\beta - z^\beta) - (q^x - z^x) \epsilon_{\alpha\beta} (\dots)}{[(\vec{p} - \vec{z}) \cdot (\vec{q} - \vec{z})]^2}$$

We know that

$$\cos \theta = (\vec{p} - \vec{z}) \cdot (\vec{q} - \vec{z}) / p^2$$

$$\frac{\partial \theta}{\partial p^x} = \frac{1}{p^4} \left[(\vec{p} - \vec{z}) \cdot (\vec{q} - \vec{z}) \epsilon_{x\beta} (q^\beta - z^\beta) - (q^x - z^x) \epsilon_{\alpha\beta} (p^\alpha - z^\alpha) (q^\beta - z^\beta) \right]$$

$$\frac{2\theta}{2p^\gamma} = \frac{1}{p^4} \left[(p^\alpha - z^\alpha)(Q^\alpha - z^\alpha) \epsilon_{\gamma\beta} (Q^\beta - z^\beta) - (p^\alpha - z^\alpha)(Q^\gamma - z^\gamma) \epsilon_{\alpha\beta} (Q^\beta - z^\beta) \right]$$

$$\frac{2\theta}{2p^\gamma} = \frac{1}{p^4} \left\{ (p^\alpha - z^\alpha) \left[(Q^\alpha - z^\alpha) \epsilon_{\gamma\beta} - (Q^\gamma - z^\gamma) \epsilon_{\alpha\beta} \right] \times (Q^\beta - z^\beta) \right\}$$

We then use the Shouten identity:

$$A^\alpha \epsilon_{\gamma\beta} + A^\gamma \epsilon_{\beta\alpha} + A^\beta \epsilon_{\alpha\gamma} = 0$$

$$A^\alpha \epsilon_{\gamma\beta} - A^\gamma \epsilon_{\alpha\beta} = A^\beta \epsilon_{\gamma\alpha} \text{ so that}$$

$$\frac{2\theta}{2p^\gamma} = \frac{1}{p^4} \left[(p^\alpha - z^\alpha)(Q^\beta - z^\beta) \epsilon_{\gamma\alpha} (Q^\beta - z^\beta) \right]$$

5.3 and since $|\vec{Q} - \vec{z}|^2 = p^2$, we have

$$\frac{2\theta}{2p^\gamma} = \frac{1}{p^2} (p^\alpha - z^\alpha) \epsilon_{\gamma\alpha}$$

and by symmetry,

$$\frac{2\theta}{2Q^\gamma} = \frac{1}{p^2} (Q^\alpha - z^\alpha) \epsilon_{\gamma\alpha}$$

5.4

Finally,

$$\frac{\partial h_a}{\partial p^\gamma} = \frac{1}{2}(1 - \cos \theta)(p^\alpha - z^\alpha)\epsilon_{\gamma\alpha}$$

$$\frac{\partial h_a}{\partial Q^\gamma} = -\frac{1}{2}(1 - \cos \theta)(Q^\alpha - z^\alpha)\epsilon_{\gamma\alpha}$$

S.S.

Next we consider $\frac{\partial h_a}{\partial z^\gamma}$ which is

$$\frac{\partial h_a}{\partial z^\gamma} = p^2(1 - \cos \theta) \frac{\partial \theta}{\partial z^\gamma} \text{ and}$$

since θ is a function of $\tilde{p}^\alpha = p^\alpha - z^\alpha$ and

$\tilde{Q} = Q^\alpha - z^\alpha$, we know that this has the property that

$$\frac{\partial \theta(\tilde{p}, \tilde{Q})}{\partial z^\gamma} = \frac{\partial \theta}{\partial \tilde{p}^\beta} \frac{\partial \tilde{p}^\beta}{\partial z^\gamma} + \frac{\partial \theta}{\partial \tilde{Q}^\beta} \frac{\partial \tilde{Q}^\beta}{\partial z^\gamma}$$

$$\frac{\partial \theta}{\partial z^\gamma} = \frac{\partial \theta}{\partial \tilde{p}^\beta} \frac{\partial \tilde{p}^\beta}{\partial z^\gamma} + \frac{\partial \theta}{\partial \tilde{Q}^\beta} \frac{\partial \tilde{Q}^\beta}{\partial z^\gamma}$$

and since $\frac{\partial \theta}{\partial p^\gamma} = \left(\frac{\partial \theta}{\partial \tilde{p}^\beta} \right) \left(\frac{\partial \tilde{p}^\beta}{\partial p^\gamma} \right)$, we

have $\frac{\partial \theta}{\partial \tilde{p}^\gamma} = \frac{\partial \theta}{\partial p^\gamma}$ thus

$$\frac{\partial \theta}{\partial z^\gamma} = \left(\frac{\partial \theta}{\partial p^\beta} \right) (-\delta_{\gamma\beta}) + \frac{\partial \theta}{\partial Q^\beta} (-\delta_{\gamma\beta})$$

$$\frac{\partial \theta}{\partial z^\gamma} = -\frac{1}{p^2} \varepsilon_{\gamma\alpha} [\tilde{p}^\alpha - \tilde{Q}^\alpha]$$

$$\frac{\partial \theta}{\partial z^\gamma} = -\frac{\varepsilon_{\gamma\alpha}}{p^2} (p^\alpha - Q^\alpha)$$

$$\boxed{\frac{\partial h_a}{\partial z^\gamma} = -\frac{1}{2}(1-\cos\theta) \varepsilon_{\gamma\alpha} (p^\alpha - Q^\alpha)} \quad 5.6$$

6 Derivatives of Features

Now we ask which sort of point we have. $\vec{p}$ could be $\vec{a}_k^\pm$ or $\vec{z}_k$ or an intersection point, $\vec{q}$. Let's start with the "easy" pieces, $\vec{a}_k^\pm$ and $\vec{z}_k$. We'll see that we can avoid dealing with $\vec{q}$ derivatives.

We review equations in section 1 so that

$$\vec{a}_k^+ = \vec{v}_k + t_k \hat{e}_k, \quad \vec{a}_k^- = \vec{v}_k - t_k e_{z'(k)}$$

$$\vec{z}_k = \vec{v}_k + c_k \hat{b}_k.$$

We have to differentiate some unit vectors.
 For a unit vector $\hat{u} = \frac{\vec{u}}{|\vec{u}|}$, we have

$$\begin{aligned}\frac{\partial \hat{u}^\alpha}{\partial u^\beta} &= \frac{\partial}{\partial u^\beta} \left(\frac{u^\alpha}{\sqrt{u^\gamma u^\gamma}} \right) \\ &= \frac{\delta^{\alpha\beta}}{|\vec{u}|} - \frac{u^\alpha u^\beta}{|\vec{u}|^3} \\ &= \frac{1}{|\vec{u}|} (\delta^{\alpha\beta} - \hat{u}^\alpha \hat{u}^\beta)\end{aligned}$$

Let $\boxed{\Pi_u^{\alpha\beta} \equiv \delta^{\alpha\beta} - \hat{u}^\alpha \hat{u}^\beta}$ 6.1

so that

$$\frac{\partial \hat{e}_j^\alpha}{\partial V_k^\beta} = \sum_r \left(\frac{\partial \hat{e}_j^\alpha}{\partial e_j^r} \right) \left(\frac{\partial e_j^r}{\partial V_k^\beta} \right)$$

$\boxed{\frac{\partial \hat{e}_j^\alpha}{\partial V_k^\beta} = \frac{1}{|\vec{e}_j|} \Pi_{e_j}^{\alpha\beta} (\delta_{z(j)k} - \delta_{jk})}$ 6.2

We also have

$$\frac{\partial \hat{b}_k^\alpha}{\partial \hat{b}_k^\beta} = \frac{1}{|\vec{b}_k|} \Pi_{b_k}^{\alpha\beta} \quad \text{thus}$$

$$\frac{\partial \hat{b}_k^\alpha}{\partial V_k^\beta} = \sum_\gamma \left(\frac{\partial \hat{b}_k^\alpha}{\partial b_k^\gamma} \right) \left(\frac{\partial b_k^\gamma}{\partial V_k^\beta} \right)$$

$$6.3 \quad \left[\frac{\partial \hat{b}_k^\alpha}{\partial V_k^\beta} = \sum_\gamma \frac{1}{|\vec{b}_k|} \Pi_{b_k}^{\alpha\gamma} \left(\frac{\partial \hat{e}_k^\gamma}{\partial V_k^\beta} - \frac{\partial \hat{e}_{z'(k)}^\gamma}{\partial V_k^\beta} \right) \right]$$

Next, now we can differentiate our points

$$\frac{\partial a_j^{+\alpha}}{\partial V_k^\beta} = \delta_{jk} \delta^{\alpha\beta} + \hat{e}_j^\alpha \left(\frac{\partial t_j}{\partial \theta_j} \right) \left(\frac{\partial \theta_j}{\partial V_k^\beta} \right) + t_j \frac{\partial \hat{e}_j^\alpha}{\partial V_k^\beta}$$

6.4

$$\frac{\partial a_j^{-\alpha}}{\partial V_k^\beta} = \delta_{jk} \delta^{\alpha\beta} - \hat{e}_{z'(j)}^\alpha \left(\frac{\partial t_j}{\partial \theta_j} \right) \left(\frac{\partial \theta_j}{\partial V_k^\beta} \right) - t_j \frac{\partial \hat{e}_{z'(j)}^\alpha}{\partial V_k^\beta}$$

6.5

Recall 1.7 and 1.8 and write

$$\frac{\partial z_j^\alpha}{\partial V_k^\beta} = \delta_{jk} \delta^{\alpha\beta} + \hat{b}_j^\alpha \left(\frac{\partial c_j}{\partial \theta_j} \frac{\partial \theta_j}{\partial V_k^\beta} \right) + c_j \frac{\partial \hat{b}_j^\alpha}{\partial V_k^\beta}$$

6.6

And we have

$$\frac{\partial t_j}{\partial \theta_j} = - \frac{\rho}{\tan^2 \theta_j / 2} \frac{\sec^2 \theta_j / 2}{2}$$

$$\frac{\partial t_j}{\partial \theta_j} = - \frac{\rho}{\sin^2 \theta_j / 2}$$

6.7

and

$$\frac{\partial C_i}{\partial \theta_i} = - \frac{\rho}{\sin^2 \theta_{k/2}} \frac{\cos \theta_{k/2}}{2}$$

$$\frac{\partial C_i}{\partial \theta_i} = - \frac{\rho}{\sin \theta_{k/2} \tan \theta_{k/2}}$$

6.8

7 Forces on the boundary

If we're considering integrating over a boundary, our area A_n is

$$A_n = \frac{1}{2} \oint_{\partial(A \cap B)} \vec{X} \times d\vec{X} \text{ for boundary } \vec{X}.$$

The variation of A_n with $\vec{X}$ is then

$$\delta A_n = \frac{1}{2} \oint_{\partial(A \cap B)} \delta \vec{X} \times d\vec{X} + \frac{1}{2} \oint_{\partial(A \cap B)} \vec{X} \times d(\delta \vec{X}).$$

Now we can perform integration by parts so

$$\oint_{\partial(A \cap B)} \vec{X} \times d(\delta \vec{X}) = \underbrace{\vec{X} \times \delta \vec{X}}_{\substack{0 \\ \Delta_0}} \Big|_{\partial(A \cap B)^-}^{\partial(A \cap B)^+} - \oint_{\partial(A \cap B)} \delta \vec{X} \times d\vec{X}$$

$$\delta A_n = \oint_{\partial(A \cap B)} \delta \vec{X} \times d\vec{X}.$$

 7.1

We can write

$$\delta \vec{X} = \sum_j \left(\frac{\partial \vec{X}}{\partial \vec{V}_j} \right) \delta \vec{V}_j \text{ so that}$$

$$\delta A_1 = \oint_{\partial(A \cap B)} \sum_j \left(\left(\frac{\partial \vec{X}}{\partial \vec{V}_j} \right) \times d\vec{X} \right) \delta \vec{V}_j$$

$$\delta A_1 = \sum_j \oint_{\partial(A \cap B)} \left(\frac{\partial \vec{X}}{\partial \vec{V}_j} \times d\vec{X} \right) \delta \vec{V}_j$$

$$\text{and because } \delta A_1 = \sum_j \frac{\partial A_1}{\partial \vec{V}_j} \delta \vec{V}_j,$$

$$\boxed{\frac{\partial A_1}{\partial V_j^\alpha} = \oint_{\partial(A \cap B)} \frac{\partial \vec{X}}{\partial \vec{V}_j} \times d\vec{X}} \quad 7.2$$

For an edge feature from $\vec{P}$ to $\vec{Q}$, we

have a specific edge,

$$\delta A_{\text{edge}} = \int_{\text{edge}} \delta \vec{X} \times d\vec{X}. \text{ If we consider}$$

$$\vec{X} = \vec{P} + \lambda(\vec{Q} - \vec{P}) \text{ then } d\vec{X} = (\vec{Q} - \vec{P})d\lambda$$

and

$$\delta A_{\text{edge}} = \int_0^1 \delta \vec{x} \times (\vec{Q} - \vec{P}) d\lambda$$

$$= \int_0^1 (\delta \vec{P} + \lambda (\delta \vec{Q} - \delta \vec{P})) \times (\vec{Q} - \vec{P}) d\lambda$$

$$= \delta \vec{P} \times (\vec{Q} - \vec{P}) + \frac{1}{2} (\delta \vec{Q} - \delta \vec{P}) \times (\vec{Q} - \vec{P})$$

$$\delta A_{\text{edge}} = \frac{1}{2} (\delta \vec{P} + \delta \vec{Q}) \times (\vec{Q} - \vec{P})$$

Now we know that if this is edge m ,

$$\text{then } \delta \vec{P} = (1 - s_p) \delta \vec{a}_m^+ + s_p \delta \vec{a}_{z(m)}^-$$

$$\delta A_{\text{edge}, m} = \frac{1}{2} \left[(1 - s_p + 1 - s_q) \delta \vec{a}_m^+ + s_p + s_q \delta \vec{a}_{z(m)}^- \right]$$

$$\delta A_{\text{edge}, m} = (1 - \bar{s}) \delta \vec{a}_m^+ + \bar{s} \delta \vec{a}_{z(m)}^-$$

$$\text{where } \bar{s} = \frac{s_p + s_q}{2}$$

7.3.

For an arc m ,

$$\vec{X}(\varphi) = \vec{Z}_m + \rho(\cos\varphi, \sin\varphi) \quad \text{So}$$

$$\delta\vec{X}(\varphi) = \delta\vec{Z}_m$$

This mean

$$\delta A_{\text{arc},m} = \delta\vec{Z}_m \times \int d\vec{X}$$

$$\boxed{\delta A_{\text{arc},m} = \delta\vec{Z}_m \times (\vec{Q} - \vec{P})} \quad 7.4$$

Then from 7.2 we have

$$\boxed{\frac{2A_\Lambda}{2V_j^\delta} = \sum_{\text{features}, m} \frac{2f_m^\alpha}{2V_j^\delta} \epsilon_{\alpha\beta} C_m^\beta} \quad 7.5$$

Where $\vec{C}_m$ is the vector representing the distance of the feature from start to end in the overlapping region boundary.

and

$$\frac{\partial f_m^\alpha}{\partial v_j^\delta} = \begin{cases} (1-\bar{s}_m) \frac{\partial \vec{a}_m^+}{\partial \vec{v}_j} + \bar{s}_m \frac{\partial \vec{a}_{\bar{z}(m)}}{\partial \vec{v}_j} & \text{for edge} \\ \frac{\partial \vec{z}_m}{\partial \vec{v}_j} & \text{for arc} \end{cases} \quad 7.6$$

8 Area and Perimeter Springs

If we want to find the area gradient of the shape itself, that's simply

$$\frac{\partial A_i}{\partial V_j^\alpha} = \sum_{k=1}^{n_i} \frac{\partial V_k^\alpha}{\partial V_j^\alpha} \epsilon_{\alpha\beta} e_k^\beta \quad \text{from}$$

equation 7.5.

Then we note $\frac{\partial V_k^\alpha}{\partial V_j^\alpha} = \delta_{kj} \delta_{\alpha\alpha}$ and so

$$\boxed{\frac{\partial A_i}{\partial V_j^\alpha} = \epsilon_{\alpha\beta} e_j^\beta} \quad 8.1$$

This is obvious, but it is a nice sanity check for eqn 7.5.

The perimeter term is simply the sum of edge lengths

$$P_i = \sum_{k=1}^{n_i} |\vec{e}_k| \quad \text{do}$$

$$\frac{\partial P_i}{\partial V_j^\alpha} = \sum_{k=1}^{n_i} \frac{(V_k^\alpha - V_{z'(k)}^\alpha)(\delta_{kj} - \delta_{z'(k)j})}{|\vec{e}_k|}$$

$$\boxed{\frac{\partial P_i}{\partial V_j^\alpha} = \frac{V_j^\alpha - V_{z'(j)}^\alpha}{|\vec{e}_j|} - \frac{V_{z(j)}^\alpha - V_j^\alpha}{|\vec{e}_{z(j)}|} \quad 8.2}$$

9 The Energy function

Let U_x be the
form every x term will have in the
energy of a packing configuration. Here
 K_x is the spring constant and $x \in \begin{cases} \text{area} \\ \text{perimeter} \\ \text{overlap} \\ \text{adhesion} \end{cases}$

We could make K_x different for every
polygon/overlap in the packing, but I don't
personally find that direction interesting.

Here $f_x \in \begin{cases} \text{area of polygon} \\ \text{perimeter of polygon} \\ \text{overlap} \\ \text{intersection distance} \end{cases}$ and
 $\bar{f}_x \in \begin{cases} \text{target area of polygon} \\ \text{target perimeter of polygon} \\ \text{target overlap} \\ \text{target intersection distance} \end{cases}$

The overlap term sets the scale and has energy

$$U_{\text{overlap}} = 2 \sum_{\text{overlaps } \{i,j\}} \left(\frac{A_{\text{overlap}}}{A_i + A_j} \right)^2 \quad 9.1$$

Note this is $\frac{1}{2} \left(\frac{A_{\text{overlap}}}{\frac{1}{2}(A_i + A_j)} \right)^2$

$$U_{\text{adhesion}} = 2 K_{\text{adh}} \sum_{\text{overlaps } \{i,j\}} \left(\frac{D_{\text{overlap}}}{\text{Span}_i + \text{Span}_j} \right)^2 \quad 9.2$$

Getting Span_i is a discussion for

later

$$U_{\text{area}} = \frac{K_A}{2} \sum_{\text{polygons } i} \left(1 - \frac{A_i}{\bar{A}_i} \right)^2 \quad 9.3$$

$$U_{\text{perimeter}} = \frac{K_P}{2} \sum_{\text{polygons } i} \left(1 - \frac{P_i}{\bar{P}_i} \right)^2 \quad 9.4$$